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Roll No

ME/IP-6005(2)-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Finite Element Method

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What are the objectives of structural analysis.
b) Differentiate between skeletal and continuum structures.
2. What do you understand by discretization? Discuss various element shapes used for discretization.
3. How do you assemble elements in finite elements method. Discuss it detail.
4. a) Explain Gaussian elimination method.
b) What are advantages of Numerical integration.
5. a) What is a natural co-ordinate system. Where it is used.
b) Differentiate between 2D rectangular and triangular requirement.

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6. How do you apply finite element method for analysis of truss?
Explain with a suitable example.
7.
 - a) Explain Hamilton's principle of finite element analysis.
 - b) How do you derive mass matrices for JD element?
8. Write short notes on any two:
 - a) Basic steps in finite element problem formulation
 - b) Axisymmetric elements
 - c) Advantages of finite element methods
 - d) Brief introduction of any FEM software
